CODE SPORT


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Over the next few columns, we will take a look at data storage systems, and how they are evolving to cater to the data-centric computing world.

Last month, we featured a special edition of 'CodeSport', which discussed the evolution of programming languages over the past 10 years, and how they are likely to evolve over the coming 10 years. The article went on to hazard a guess that the 'Big Data' explosion would shift the momentum to languages that make data processing simple and efficient, and make programs 'data-centric' instead of the 'code-centric' perspective. Many of our readers had responded with their own views on how they see computing paradigms evolving over the coming 10 years. Thanks a lot to all our readers for their feedback and thoughts.

One of our readers, Ravikrishnan, sent me a pertinent comment, which I want to share: "Thank you for your article on the evolution of programming languages. Indeed, there is a heavy momentum towards processing huge amounts of data using commodity hardware and software. While the basic concepts and algorithms of computer science would continue to hold sway, the sheer scale of the data explosion would require programmers to understand and apply algorithms where data does not fit in main memory. Hence programmers need to start worrying about data latency of secondary storage such as flash SSD/disk storage systems. In a way, the shift towards data-centric computing means more intelligent storage systems, and a need for programmers to understand about state-of-the-art storage systems, where big data is stored, processed and preserved. While this is not a traditional topic covered in 'CodeSport', given the importance of data storage in a 'Big Data' world, it would be great if 'CodeSport' does a deep-dive into state-of-the-art storage systems in a future column."

It was a timely reminder for me. While I have discussed various 'Big Data' computing paradigms in some of our past columns, I have not covered storage systems at all. So over the next few columns, I am going to discuss storage systems, and how they have evolved over years to cater to the 'Big Data' explosion. I will take readers through some of the challenging problems as well as the state-of-the-art research directions in this space.

A storage systems primer

Let us start our journey into storage by understanding some of the basic concepts and terminology. In the traditional view of storage, we all know about the triumvirate of the CPU, memory and disk, where the hard disk (also known as secondary storage) is part of, or directly attached to your computer, and acts as the permanent storage. From now onwards, when I use the term 'storage', I actually imply the traditional secondary storage, which acts as the backup to the main memory (which is the primary storage). These include hard disk drives, flash/SSD storage, tape drives, etc.

Traditional HDDs are accessed using a variety of protocols such as SCSI, ATA, SATA and SAS. SCSI stands for Small Computer System Interface and is a parallel peripheral interface standard widely used in personal computers for attaching printers and hard-disks. ATA is another interface used for attaching disks; also known as IDE, wherein the controller is integrated into the disk drive itself, ATA is also a parallel interface like SCSI and both have their equivalent serial interfaces namely, Serial SCSI (abbreviated as SAS) and Serial ATA (abbreviated as SATA) which allow a serial stream of data to be transmitted between the PC and the disk drive.

Note, however, that in the traditional view of storage, it is part of the compute server, since it is directly attached to the server and is accessed through